EDA Report: Travel Planning Datasets

AI Travel Planning Assistant, Agentic AI Capstone

1. Dataset Overview

The project uses three JSON datasets stored in the data/ folder. Together they cover 8 Indian cities: Delhi, Mumbai, Goa, Bangalore, Chennai, Hyderabad, Kolkata, and Jaipur.

Dataset	Records	Columns	Purpose
flights.json	30	7	Flight search, cheapest and fastest selection
hotels.json	40	6	Hotel recommendation and budget
places.json	40	5	Attraction discovery and itinerary

2. Data Quality

Missing values: None in any dataset. All 18 columns are complete.

Duplicates: No duplicate IDs (flight_id, hotel_id, place_id). Every record is unique.

Data quality issues found:

1. Place names are generic and repeated (for example, Famous Fort, Scenic Museum, Historic Park). Three names repeat within the same city.

2. Place type and name do not match. For example, Famous Fort in Delhi is typed as lake, and Scenic Museum in Kolkata is typed as temple.

3. Flight dates are all from 2025 (4 Jan to 27 Dec 2025). The app uses the dataset schedule on the user's date, which is documented as a limitation.

4. Flight durations are coarse: only 60, 75, 150, and 240 minutes. Real routes have longer durations, so the data is simplified.

5. Stars and price have almost no relationship (correlation 0.072). A 2-star hotel can cost more than a 5-star hotel.

6. Goa has no beach type in places.json, which is why the itinerary uses other types when beach is requested.

3. Key Statistics

Flights

Metric	Value
Price (Rs), mean	5,511
Price (Rs), median	5,422
Price (Rs), range	2,792 to 8,981
Duration (minutes), median	150
Unique routes	25 (5 routes have more than one flight)
Airlines	Air India (10), SpiceJet (8), IndiGo (6), Go First (5), Vistara (1)

Hotels

Metric	Value
Price per night (Rs), mean	4,130
Price per night (Rs), median	4,070
Price per night (Rs), range	1,232 to 6,481
Stars distribution	2-star (14), 3-star (6), 4-star (9), 5-star (11)
Hotels with 4+ stars	20 (11 of them at or under Rs 5,000)
Stars and price correlation	0.072

Places

Metric	Value
Rating, mean / median	4.18 / 4.20
Rating range	3.50 to 4.90
Places rated 4.0 or above	28 out of 40 (70%)
Types	park (8), market (7), museum (6), temple (5), lake (4), fort (4), beach (4), monument (2)

4. Insights

1. Hotel price is not driven by stars, so the app ranks hotels by stars first and then by price. This makes a cheap 5-star hotel (Royal Heritage, Rs 1,232) a top pick.

2. Four-star options are limited. Only 11 hotels with 4+ stars cost Rs 5,000 or less, so a tight budget with a high star requirement can return no results. The app shows the closest alternatives in that case.

3. Interest-based itineraries are constrained. Some cities have few places of a given type, so the itinerary falls back to top-rated places and shows a message.

4. Flight selection is simple. Most routes have one flight, so cheapest and fastest often return the same option.

5. Limitations

- Sample datasets, not live data. Prices and availability do not reflect the real market.

- Flight dates are historical, so the schedule is an estimate for future dates.

- Place names and types can conflict, so the itinerary shows the type but not a description.

- Small size (30 to 40 records per dataset) limits the statistical power of any analysis.

6. Next Steps

- Replace sample data with a real source (scraped or API-based hotel and flight prices).

- Add place descriptions and coordinates for better itineraries and map views.

- Normalize place names and types so that they match each other.

- Add a validation step when loading data, with unique IDs per city.